

ABDUL RAZAQ

RESEARCHER IN MULTILINGUAL AND LOW-RESOURCE NLP

Hangu, Khyber Pakhtunkhwa, Pakistan | innovativerazaq.it@gmail.com | +92-332-885-5821

Portfolio: razaqnl.github.io | **GitHub:** github.com/razaqnl | **LinkedIn:** linkedin.com/in/abdul-razaq-656b3331b

RESEARCH PROFILE

B.S. Information Technology graduate from the University of Agriculture, Peshawar (CGPA 3.36/4.00), working at the intersection of multilingual NLP, linguistic data science, and applied machine learning. Current research focuses on Romanized and code-mixed Pashto: building language resources, studying annotation and data quality, evaluating multilingual models, and testing how systems generalize between human-written and AI-generated data. A manuscript on Romanized Pashto sentiment classification is in preparation.

RESEARCH INTERESTS

Multilingual and low-resource NLP	Romanized and code-mixed language
Pashto NLP and language resources	Dataset construction and annotation
Model evaluation and generalization	LLMs, RAG, and AI for education

RESEARCH EXPERIENCE AND OUTPUTS

Sentiment Analysis of Romanized Pashto on Social Media *Final Year Project, University of Agriculture, Peshawar | 2025-2026*

- Constructed a 5,339-sentence manually annotated dataset of Romanized, code-mixed Pashto from social media sources using Positive, Negative, and Neutral labels.
- Fine-tuned XLM-RoBERTa and compared it with SVM, Random Forest, Logistic Regression, and Naive Bayes using weighted F1-score evaluation.
- Identified and corrected a data-leakage issue in an earlier split, improving the reliability and reproducibility of reported evaluation.
- Manuscript under preparation; source code and dataset are available through the research project repository.

Current Research Experiments *Romanized Pashto research programme | Ongoing*

- AI-generated versus human-written data: comparing distributions, class balance, and dataset quality.
- Cross-source generalization: training on AI-generated data and testing on human-written data, then reversing the direction.
- Investigating how data composition, tokenization, representation, and multilingual transfer influence model behavior.

SELECTED RESEARCH AND TECHNICAL PROJECTS

Pashto Sentiment Annotation Platform *FastAPI | Annotation infrastructure | Ongoing*

- Built a multi-role annotation and review platform to extend and quality-control the Romanized Pashto sentiment dataset.

Retrieval-Augmented Generation (RAG) Assistant *Python, LangChain, FastAPI, Hugging Face, Groq, Pinecone*

- Developed an end-to-end retrieval pipeline with document ingestion, semantic search, embeddings, and LLM response generation for a portfolio knowledge base.

AI Resume Analyzer *Python, FastAPI, Groq API*

- Built an API that extracts structured information from PDF and DOCX resumes and generates role-specific feedback.

Pashtology Language Testing Platform *Language assessment | Deployed on Netlify*

- Built and deployed a language testing platform for students as an applied language-technology project.

EDUCATION

Bachelor of Science in Information Technology *University of Agriculture, Peshawar | Graduated July 2026 | CGPA: 3.36/4.00*

Relevant coursework: Machine Learning, Natural Language Processing, Database Systems, Data Structures, and Software Engineering.

METHODS AND TECHNICAL SKILLS

Programming and data: Python, SQL, JavaScript, Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn, EDA, statistical analysis, web scraping, and data processing.

NLP and AI: XLM-RoBERTa, Hugging Face Transformers, mT5, tokenization, text classification, LLMs, RAG, embeddings, vector databases, prompt engineering, and AI agents.

Research engineering: FastAPI, REST APIs, JWT authentication, PostgreSQL, MySQL, SQLite, Git, GitHub, Docker, LangChain, PyTorch, and experiment evaluation.

PUBLICATIONS AND RESEARCH LINKS

Manuscript in preparation: Romanized Pashto sentiment classification and evaluation. No peer-reviewed publications yet.

Research repository: github.com/razaqnlp/Romanized_pashto_research

Model resource: huggingface.co/abdul-razaq-ork/Romanized_Pashto_Sentiment

CERTIFICATIONS

Machine Learning Specialization - DeepLearning.AI and Stanford University; Deep Learning Specialization - DeepLearning.AI; Microsoft Azure AI Fundamentals (AI-900); Python for Everybody Specialization - University of Michigan; Data Analysis with Python - IBM; Certified in Artificial Intelligence - NAVTTC.

LANGUAGES

Pashto (Native) | Urdu (Native) | English (Professional)

Abdul Razaq | Research CV | 2026